

भारतीय मानक
Indian Standard

IS 17803 : 2022

**पेयजल की बोतल
(कॉपर, स्टेनलेस स्टील, अल्युमिनियम) —
विशिष्टि**

**Potable Water Bottles (Copper,
Stainless Steel, Aluminium) —
Specification**

ICS 55.100; 67.250

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February 2022

Price Group 4

Utensils, Cutlery Domestic Hardware Sectional Committee, MED 33

FOREWORD

This Indian Standard was adopted by the Bureau of Indian Standards, after the draft finalized by the Utensils, Cutlery and Domestic Hardware Sectional Committee had been approved by the Mechanical Engineering Divisional Council.

This standard covers potable copper, stainless steel and aluminium bottle used for carrying of drinking water.

This standard does not cover insulated and vacuum flask. Vacuum insulated flask is covered in IS 17526 : 2021 'Stainless steel vacuum flasks — Specification'. Insulated type of flask is covered in IS 17790 : 2022 'Insulated flask for domestic use — Specification'.

While preparing this standard, assistance has been taken from BS EN 12546-1 : 2000 'Materials and articles in contact with foodstuffs. Insulated containers for domestic use: Part 1 Specification for vacuum ware, insulated flasks and jugs'. While implementing this standard, the manufacturer shall ensure compliance with relevant statutory regulations.

The relevant SI units and corresponding conversion factors are given below for guidance:

Pressure 1Pa (Pascal)	=	1 N/m ²
1 kgf/mm ²	=	9.806 65 MPa

The composition of the committee responsible for the formulation of this standard is given in Annex A.

For the purpose of deciding whether a particular requirement of this standard is complied with the final value, observed or calculated, expressing the result of a test or analysis shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Indian Standard

POTABLE WATER BOTTLES (COPPER, STAINLESS STEEL, ALUMINIUM) — SPECIFICATION

1 SCOPE

1.1 This standard covers minimum requirements of potable copper, stainless steel, aluminium bottle for carrying drinking water up to 1 500 ml.

1.2 This Standard cover the addition requirements and shapes of aluminium bottle. Basic requirements of aluminium bottle are covered in IS 3603.

1.3 This standard does not cover insulated and vacuum flask.

2 REFERENCES

The standard listed below contain provision which, through reference in this text, constitute provision of this standard. At the time of publication, edition was valid. All standards are subject to revision and parties to agreements based on this standard are encouraged to investigate possibility of applying the most recent edition of the standard indicated below:

<i>IS No</i>	<i>Title</i>
191 : 2007	Copper — Specification (fourth revision)
1868 : 1996	Anodic coatings on aluminium and its alloys — Specification (third revision)
2471 : 1963	Methods of test for metal containers
3565 : 2018	Teats for feeding bottles — Specification (first revision)
3603 : 1988	Specification for seamless aluminium bottles (first revision)
5522 : 2014	Stainless steel sheets and strips for utensils — Specification (third revision)
5856 : 2017	Welding consumables — Wire electrodes, strip electrodes, wires and rods for arc welding of stainless and heat resisting steels — Classification (second revision)
6057 : 1988	Specification for hard anodic coatings on aluminium and aluminium alloys (first revision)

IS No

Title

6911 : 2017	Stainless steel plate, sheet and strip — Specification (second revision)
8688 : 1988	Specification for plastics potable water bottles (first revision)
9730 : 2008	Non-stick unreinforced plastics coatings on domestic cooking utensils — Specification (first revision)
9806 : 2001	Methods of test for and permissible limits of toxic materials released from ceramic ware, vitreous enamelware, glassware and glass-ceramic ware in contact with food (first revision)
9845 : 1998	Determination of overall migration of constituents of plastics materials and articles intended to come in contact with foodstuffs — Method of analysis (second revision)
10500 : 2012	Drinking water — Specification (second revision)

3 TERMINOLOGY

For the purpose of this standard, the following definitions shall apply.

3.1 Domestic Copper Bottle — Potable bottle made of food grade copper, for carrying drinking water.

3.2 Domestic Stainless Steel Bottle — Potable bottle made of food grade stainless steel for carrying drinking water.

3.3 Domestic Aluminium Bottle — Potable bottle made of treated aluminium for carrying drinking water.

NOTE — Refer 7.1.8 for treated aluminium anodic coating or hard anodised coating.

3.4 Nominal Capacity — Volume of water at room temperature required to fill the bottle, when in upright position, to a level 10 mm below the lowest part of the inserted/screwed closure.

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4 TYPES (SHAPE, SIZE AND NOMINAL CAPACITY)

4.1 The bottle normally comes in various capacities upto 1 500 ml. The nominal capacity of the bottle shall be declared by the manufacturer.

4.2 The shape, size and nominal capacity of the bottle shall be as agreed to between the purchaser and the manufacturer.

5 MANUFACTURE AND WORKMANSHIP

5.1 The bottle shall be clean, reasonably free from distortion, dents, wrinkles, wavy surface, colouring, burrs, scratches, pitting, deep tool marks and other surface defects. The design of the bottle shall be such that it is easy to clean and prevent accumulation of dirt.

5.2 The bottle shall be subjected to tests as specified in 7.

5.3 If possible, avoid welding and have only seamless stainless bottle. Welding if done for stainless steel bottle, shall be free from welding defects. The welding electrode shall conform to grade 308 or higher grade as per IS 5856. The weld shall be finished properly so that when a finger is passed on the joint, it shall be smooth.

5.4 The mouth of the bottle which would be used for sipping shall have smooth, no sharp edges.

5.5 Bottles shall be seamless drawn to avoid welding on copper or aluminium.

6 SAMPLING

For the purpose of conducting the tests, 3 samples of the bottle of each individual size shall be tested. In the event of one of the 3 samples failing the test, additional 7 samples shall be tested and these shall all pass.

7 PERFORMANCE TESTS

7.1 Materials

7.1.1 Copper Bottle

7.1.1.1 The copper sheet used for drawing of bottle shall comply to the chemical properties as per Cu-ETP, Cu-FRHC or Cu-CATH-2 given in Table 4 of IS 191. The minimum sheet thickness of bottle shall be 0.5 mm. The minimum sheet thickness of lid and cap shall be 0.5 mm. The nominal thickness measured should the thickness at the place where little or no work has been done, at the base.

7.1.2 Stainless Steel Bottle

7.1.2.1 The material used to manufacture shall be of grade 304 as per IS 5522 or grade 316L as per IS 6911. The minimum sheet thickness of bottle shall be

0.4 mm. The minimum sheet thickness of lid and cap shall be 0.4 mm. The nominal thickness measured should the thickness at the place where little or no work has been done, at the base.

7.1.3 Stopper

7.1.3.1 The stopper of the bottle shall be of non-toxic and food grade material (*see* IS 9845). If stainless steel, copper or aluminium is used, stopper shall conform to the material specified for the respective bottle. When fitted to the bottle it shall not slip off and shall provide a leak proof fitting.

7.1.3.2 If an O-ring or washer is provided it shall be food grade silicone, refer to IS 3565 for information on silicone. Silicone manufacturers certificate for food grade is acceptable. Only physical and chemical properties needs to be complied with for silicone O-ring or washer.

7.1.3.3 The O-ring or washer, when visually examined, shall be free from blisters, pinholes, cracks, embedded foreign matters and other defects which may impair their serviceability. No joints allowed

7.1.4 Auxiliary Closure

The auxiliary closure(s) can be made of polymer. It shall be non-toxic, food grade material and shall not deform under normal conditions of use (*see* to IS 9845 for compliance). The materials shall not contain bisphenol A (BPA). If auxiliary closure(s) is made of stainless steel, copper or treated aluminium, the material shall be same as that used for the bottle

7.1.5 Lever, device for operating or regulating flow (if provided), should be made of food grade material. Supplier certificate for food grade is acceptable. The lever, device for operating or regulating of flow shall be operated for a minimum of 1 500 times. It shall be checked for proper functioning.

7.1.6 Shoulder strap or cord to hang the bottle on a hook, if provided, shall comply with the respective requirements given in 7.4 or 7.5.

7.1.7 Cord chain, if provided to hold the stopper/cap along with the bottle shall comply with the requirement as given in 7.6.

7.1.8 The aluminium bottle, if treated with anodic coating shall comply to IS 1868. The minimum thickness of the anodic coating shall be 10 micron. The aluminium bottle, if hard anodised, shall comply to IS 6057. The minimum thickness of the hard anodized coating shall be 25 micron.

7.1.9 The aluminium bottle if coated fully or partially externally, shall conform to the thickness test, salt water corrosion test, adhesion test procedures as per IS 9730. The limit for lead and cadmium of the coating

shall be as per IS 9806 (*see* no iv and v of Table 1 IS 9806), when measured as per standard test method. The minimum thickness of the coating shall be declared by the manufacturer.

7.2 Capacity Test

7.2.1 The nominal capacity of the bottle shall be declared amongst up to 1 500 ml. Bottles with other capacities may be manufactured as agreed to between the purchaser and the manufacturer. The tolerance on the nominal capacity shall be + 5 percent. No negative tolerance is permitted. The capacity of the bottle shall be checked as per **7.2.2**.

7.2.2 Ensure the bottle in empty. Take water in a calibrated jar. Pour the water from the calibrated jar into the bottle to a level 10 mm below the lowest part of the inserted closure. The difference in reading of water level in the calibrated jar is the capacity of the bottle.

7.3 Impact Resistance Test

7.3.1 Drop Impact Test

7.3.1.1 Fill water in the bottle up to its full capacity at the normal temperature and drop it horizontally from the height of 400 mm in a vertical state on a hard wood board of 30 mm or more in thickness fixed.

7.3.1.2 The bottle shall be free from defects which are harmful to use. There shall be no crack or water leakage from the bottle.

7.3.2 Pendulum Impact Test

7.3.2.1 Fill water in the bottle up to its full capacity at the normal temperature. Raise the bottle to the position of 45° by making the length of the shoulder strap 400 mm. Make the bottle collide with a hard wood board of 30 mm or more in thickness fixed vertically. Typical arrangement for test is shown in Fig. 1.

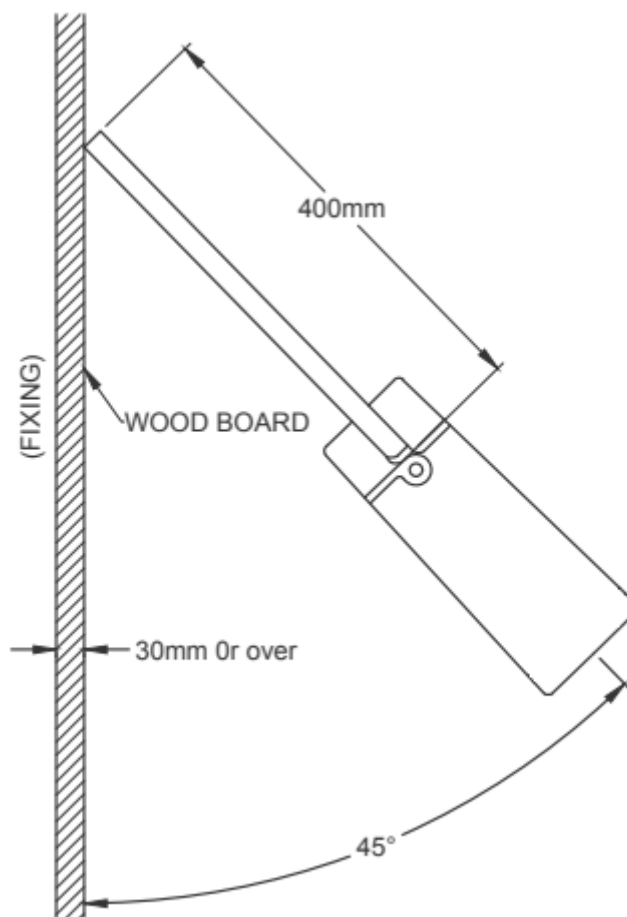


FIG. 1 PENDULUM IMPACT TEST

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7.3.2.2 The bottle shall be free from defects which are harmful to use. There shall be no crack or water leakage. The bottle shall comply to leakage test as specified in 7.7.

7.4 Test for Strength of Shoulder Strap

7.4.1 A shoulder strap if provided, shall be subjected to the test as given in 7.4.1.1.

7.4.1.1 Weigh the bottle with water filled to its maximum capacity along with the stopper(s) and auxiliary(s) as applicable. Apply a load, equal to a minimum of ten times its weight, typical arrangement as shown in Fig 2. After the maximum load is attained, keep for 5 min and examine the presence of any abnormality.

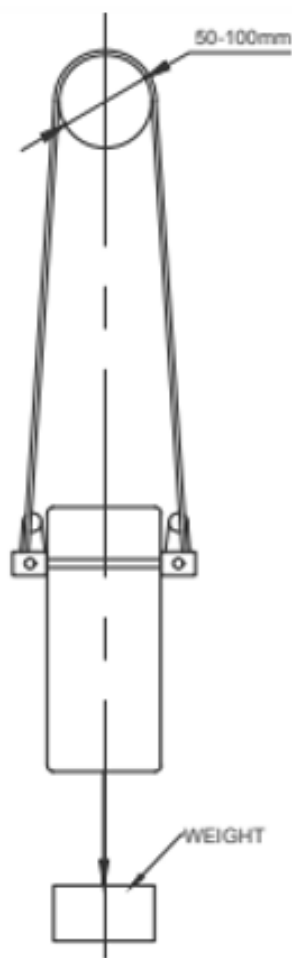


FIG. 2 TEST FOR THE SHOULDER STRAP

7.4.2 After or during this test, the strap shall not break. The area on the bottle to which the strap is fastened shall not show any signs of damage or defect. The elongation under load shall not exceed 10 percent of the total length of the strap. Initial stretching due to unwinding of the strap shall not be taken as elongation for the purpose of calculation.

NOTE — The material of the strap may be flexible and shall not be affected by water.

7.5 Test for Cord

7.5.1 In case instead of a shoulder strap if a cord is provided to tag the bottle to a hook, the strength of the cord shall be subjected to the test as given in 7.5.1.2.

7.5.1.2 Take a random sample other than that specified in 6. Put small steel balls equivalent to 2.5 times the weight of water that the bottle can hold. Hang the filled bottle by a suitable means. Ensure the bottle is hanging freely without touching any surface. Check the strap after 24 h, there shall be no visual defects or physical damage. The area on the bottle to which the strap is fastened shall not show any sign of damage or defect.

7.6 Test for Cord Chain

Take a random sample other than that specified in 6. Put small steel balls equivalent to 1.5 times the weight of water that bottle can hold. Hang the filled bottle by a suitable means. Ensure the bottle is hanging freely without touching any surface. Check the cord chain after 5 min, there shall be no visual defects or physical damage. The area on the bottle/cap to which the cord chain is fastened, shall not show any signs of damage or defect.

7.7 Leakage Test

7.7.1 Fill the bottle to 75 percent of its nominal capacity with water at 95 °C containing 0.5 percent surfactant. Close the stopper with a torque of 2 N.m or if not fitted with a screwed stopper, push in the stopper to its full extent. Thoroughly dry the outside of the stopper, sipper etc. Put the bottle upside down position for at least 10 min. No drops shall appear on the stopper, sipper or on the surface on which the bottle has been over turned. The bottle shall be tested without the auxiliary cover, if the design of the bottle is provided with an auxiliary cover along with the stopper.

7.7.2 The bottle shall be subjected to internal air pressure of 50 kPa (around 0.5 kgf/cm²) for 15 s, shall not show any sign of leakage when immersed in water. The test shall be carried out as prescribed in 3 of IS 2471.

7.8 Staining Test (Stainless Steel Bottle Only)

7.8.1 The inner and outer surface of the bottle shall be thoroughly washed with hot soapy water. Thoroughly rinse and then degrease the test specimens in acetone or methylated spirits, then wiped using a soft cloth. The bottle, when dipped for 16 h in each of the following solutions maintained at 60 ± 2 °C temperature, shall not show any sign of staining after removal from the solutions at the end of above period:

- Ten gram of glacial acetic acid (99 percent) dissolved in distilled water to make 100 ml; and
- Five gram of pure sodium chloride dissolved in distilled water to make 100 ml.

7.9 Lever, Device for Operating or Regulating Flow Test

Operate the lever or device 1 500 times and check for proper functioning of its intended use as per the instructions of manufacturer.

7.10 Test for Stability of Bottle

When the bottle filled with water to the full capacity is placed upright over a rough 10° inclined plane, it should not overturn or fall.

7.11 Load Test

Empty bottle in lid close condition apply the load test. Load of 5 kg shall be applied vertically on bottle surface for 30 min. There shall not be any deformation in shape/bulging.

7.12 If any polymer material is used which comes in contact with the stored drinking water shall be food grade, shall comply to food safe test/leaching test as per IS 9845. The polymer used shall be BPA free. Supplier certificate is acceptable. The bottle shall not degrade the drinking water, the product shall comply with the test as per 5 of IS 8688.

7.13 The bottle shall be properly rinsed with RO or distilled water. The bottle then shall be filled to capacity with RO or distilled water. Let it store for 24 h. The water when tested after storing for minimum 24 h, shall comply with the physical requirements of IS 10500.

8 INSTRUCTIONS FOR USE

8.1 Bottle shall carry adequate instructions to the user for its proper use to ensure satisfactory performance.

8.2 The manufacturer shall provide the following minimum information along with each bottle as given below:

- a) How to unpack and take care of environment;

- b) How to stock the bottle in case the bottle is not going to be used, to avoid pitting or getting contaminated;
- c) Wash thoroughly before and after use;
- d) Always rinse with clean water before each use;
- e) Ensure the bottle is cleaned, empty and dry before stocking after each use;
- f) Do not use for carrying ice cubes; do not put in freezer.
- g) Do not use to store or carry carbonated drinks;
- h) The bottle should be periodically cleaned;
- j) Never use in a dishwasher, unless otherwise stated by the manufacturer; and
- k) Keep these instructions safe.

9 MARKING

9.1 Each bottle shall be legibly and permanently marked with the following minimum information:

- a) Nominal capacity;
- b) Grade of material of bottle along with applicable IS No.;
- c) Name or trade mark of the manufacturer; and
- d) Batch number.

9.2 BIS Certification Marking

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act, 2016* and the Rules and Regulations framed thereunder, and the product(s) may be marked with the Standard Mark.

10 PACKAGING

Each bottle shall be suitable packaged to avoid any damage during transit. The packaging shall be eco-friendly.

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ANNEX A

(Foreword)

COMMITTEE COMPOSITION

Utensils, Cutlery Domestic Hardware Sectional Committee, MED 33

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Amendments are issued to standards as the need arises on the basis of comments. Standards are also reviewed periodically; a standard along with amendments is reaffirmed when such review indicates that no changes are needed; if the review indicates that changes are needed, it is taken up for revision. Users of Indian Standards should ascertain that they are in possession of the latest amendments or edition by referring to the latest issue of 'BIS Catalogue' and 'Standards: Monthly Additions'.

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Amendments Issued Since Publication

Amend No.	Date of Issue	Text Affected

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